B.Tech Biotechnology
Sem VI

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UNIT -4

Baking and Biosensors

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Biotechnology of Baking

Dough Rheology

Dough rheology is critical for understanding the behavior of dough during mixing, fermentation, and baking. The rheological properties of dough affect the texture, volume, and overall quality of the final baked product.

1. **Viscoelastic Properties**:
    1. Dough behaves as both a viscous liquid and an elastic solid. This viscoelastic behavior is essential for gas retention and dough stability during fermentation and baking.
    2. **Elasticity**: The dough's ability to stretch and recover its shape, primarily due to the gluten network.
    3. **Viscosity**: The dough's resistance to flow, influenced by the hydration level and the presence of other ingredients.

2. Gluten Network:

•Gluten is a protein complex formed when wheat flour is mixed with water. It consists of two main proteins: glutenin and gliadin.

•**Glutenin**: Contributes to dough strength and elasticity.

•**Gliadin**: Provides extensibility and allows the dough to stretch.

•Proper gluten development is achieved through kneading, which aligns and strengthens the gluten strands, creating a network that traps gas produced during fermentation.

3. Hydration:

•Water content in the dough affects its consistency and extensibility. Higher hydration levels result in a softer, more extensible dough, while lower hydration leads to a stiffer dough.

•Optimal hydration ensures proper gluten development and dough handling characteristics.

Enzymes:

- Enzymes like amylases and proteases are naturally present in flour or can be added during dough preparation.
- **Amylases**: Break down starches into fermentable sugars, providing food for yeast and enhancing dough fermentation.
- **Proteases**: Modify gluten structure by breaking down protein bonds, improving dough extensibility and reducing dough resistance.

Preparation of Baked Goods

1. **Breads**:
   Ingredients: Flour, water, yeast, salt, and sometimes sugar.
   Process:
   1. **Mixing**: Combine ingredients to form a cohesive dough.
   2. **Kneading**: Develop gluten network by folding and stretching the dough.
   3. **Fermentation**: Allow the dough to rise, during which yeast ferments sugars, producing carbon dioxide and ethanol.
   4. **Shaping**: Form the dough into desired shapes (e.g., loaves, rolls).
   5. **Proofing**: A final rise before baking, enhancing dough volume and texture.
   6. **Baking**: Bake at the appropriate temperature to achieve a golden crust and fully cooked interior.

Biscuits:

- **Ingredients**: Flour, fat (butter or shortening), baking powder, milk.
- **Process**:
  - **Mixing**: Combine dry ingredients, then cut in fat until the mixture resembles coarse crumbs.
  - **Hydration**: Add milk to form a soft dough.
  - **Shaping**: Roll out the dough and cut into shapes.
  - **Baking**: Bake at a high temperature until golden brown and flaky.

Crackers:

- **Ingredients**: Flour, water, salt, and optional flavorings.
- **Process**:
  - **Mixing**: Combine ingredients to form a stiff dough.
  - **Rolling**: Roll the dough thinly.
  - **Cutting**: Cut into desired shapes (e.g., squares, circles).
  - **Baking**: Bake at a high temperature until crisp and golden.

Cakes:

•**Ingredients**: Flour, sugar, eggs, butter, leavening agents (baking powder or baking soda), and flavorings.

•**Process**:

- **Creaming**: Beat butter and sugar until light and fluffy.
- **Mixing**: Add eggs, flour, and leavening agents, alternating with liquids (e.g., milk).
- **Baking**: Pour batter into pans and bake at a moderate temperature until a toothpick inserted comes out clean.

Buns:

- **Ingredients**: Flour, water, yeast, salt, and enriching ingredients (milk, butter, sugar).
- **Process**:
  - **Mixing**: Combine ingredients to form a soft dough.
  - **Kneading**: Develop gluten network.
  - **Fermentation**: Allow dough to rise.
  - **Shaping**: Form into individual bun shapes.
  - **Proofing**: Allow a final rise.
  - **Baking**: Bake until golden brown.

Pizza:

- **Ingredients**: Flour, water, yeast, salt, olive oil.
- **Process**:
  - **Mixing**: Combine ingredients to form a cohesive dough.
  - **Kneading**: Develop gluten network.
  - **Fermentation**: Allow dough to rise.
  - **Shaping**: Roll out the dough into a flat shape.
  - **Topping**: Add desired toppings (e.g., tomato sauce, cheese, vegetables, meats).
  - **Baking**: Bake at a high temperature until the crust is crispy and toppings are cooked.

Innovative Applications in Baking

1. **Gluten-Free Products**:
    1. Use of alternative flours (e.g., rice, almond, coconut) and hydrocolloids (e.g., xanthan gum, guar gum) to mimic the viscoelastic properties of gluten.
    2. Enzymes and protein sources (e.g., amylases, proteases, soy protein) enhance dough structure and texture.
2. **Sourdough Fermentation**:
    1. Natural fermentation using wild yeasts and lactic acid bacteria improves flavor, texture, and nutritional value.
    2. Long fermentation times enhance the digestibility and shelf life of sourdough bread.

3. Enzyme Technology:

1. Specific enzymes (e.g., xylanase, lipase) improve dough handling, extend shelf life, and enhance nutritional profiles.
2. Enzyme blends tailored for specific applications, such as reducing sugar content while maintaining sweetness.

- These advancements and detailed processes highlight the critical role of biotechnology in the baking industry, leading to the creation of high-quality, diverse, and innovative baked products.

BIOSENSORS

General Aspects

- **Biosensors** are analytical devices that combine a biological recognition element with a physicochemical transducer to detect and measure specific substances. They are used in various fields, including healthcare, environmental monitoring, and food analysis. The key components of a biosensor include:

1. **Biological Recognition Element**: This can be enzymes, antibodies, nucleic acids, or whole cells that specifically interact with the target analyte.
2. **Transducer**: Converts the biological interaction into a measurable signal. Types of transducers include electrochemical, optical, thermal, and piezoelectric.
3. **Signal Processor**: Amplifies and processes the signal, converting it into a readable output.

Biosensors for Food Component Analysis

Biosensors are increasingly used for the analysis of food components, providing rapid and accurate measurements of nutrients, additives, and other important substances.

1. **Nutrient Analysis**:
    1. **Glucose Biosensors**: Used to measure glucose levels in beverages and food products. They typically use glucose oxidase as the biological element and an electrochemical transducer.
    2. **Lactose Biosensors**: Measure lactose in dairy products, beneficial for lactose-intolerant consumers. These often use lactase enzymes.

2. Protein and Vitamin Analysis:

- **Protein Biosensors**: Detect specific proteins or allergens in food products. Immunosensors, which use antibodies, are commonly used for protein detection.
- **Vitamin Biosensors**: Measure vitamins like vitamin C and B12 in food products. Enzymatic biosensors are often used for this purpose.

Biosensors for Food Contaminant Analysis

Biosensors play a critical role in detecting contaminants in food, ensuring safety and compliance with regulatory standards.

1. **Pesticide Detection**:
    1. **Organophosphate Biosensors**: Detect organophosphate pesticides in fruits and vegetables. These biosensors use enzymes like acetylcholinesterase, which interact with the pesticides, and an electrochemical transducer to measure the inhibition of enzyme activity.
2. **Heavy Metal Detection**:
    1. **Lead and Cadmium Biosensors**: Used to detect heavy metals in food products. These biosensors often use metallothioneins or phytochelatins as the biological element and an electrochemical transducer.

3. Antibiotic Detection:

•**Beta-lactam Antibiotic Biosensors**: Measure residues of beta-lactam antibiotics in milk and meat products. Enzyme-based biosensors are commonly used for this analysis.

4. Toxin Detection:

•**Aflatoxin Biosensors**: Detect aflatoxins in grains and nuts. Immunosensors using specific antibodies for aflatoxins are widely used.

Commercially Available Biosensors for Food Analysis

Several biosensors are commercially available and widely used in the food industry for quality control and safety assurance.

1. **Glucose Biosensors**:
    1. **Accu-Chek**: A widely used glucose biosensor for measuring glucose levels in various food products.
    2. **Biosensor**: Employs glucose oxidase enzyme and an electrochemical transducer.

2. **Pathogen Detection Biosensors**:
    1. **iQ-Check**: A PCR-based pathogen detection system used to identify harmful microorganisms like Salmonella and E. coli in food products.
    2. **Veratox**: An ELISA-based biosensor for detecting allergens and pathogens in food.

3. Pesticide Detection Biosensors:

•**Neogen Reveal**: A lateral flow assay biosensor for detecting pesticide residues in fruits and vegetables.

•**Charm EZ**: A biosensor system for detecting various contaminants, including pesticides, in food samples.

4. Allergen Detection Biosensors:

•**AllerFlow Gluten**: A lateral flow device for detecting gluten in food products.

•**R-Biopharm RIDASCREEN**: An ELISA-based biosensor for detecting multiple food allergens.

Conclusion

- Biosensors offer a rapid, sensitive, and specific method for analyzing food components and detecting contaminants.
- They play a crucial role in ensuring the safety and quality of food products.
- The continued development and commercialization of biosensors are likely to enhance food safety monitoring and compliance with regulatory standards.

THANK YOU